Rishabh Sarma | Curriculum Vitae

Department of Mathematics, 309 McAllister Building
The Pennsylvania State University, University Park, PA-16802, US
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Education

The Pennsylvania State University — **State College, United States**
S. Chowla Research Assistant Professor — *2025-present*
Mentor : Ae Ja Yee

University of Florida — **Gainesville, United States**
PhD in Mathematics — *2019-2025*
GPA – 4.00/4.00
Thesis : Modular symmetries for rank-type partition statistics
Advisor : Frank Garvan

Université Paris Cité — **Paris, France**
Chateaubriand Fellow — *Fall 2023 (Aug-Dec)*
Advisor : Jeremy Lovejoy

Indian Institute of Technology Guwahati — **India**
Master of Science in Mathematics and Computing — *2016-2018*
Masters thesis : Construction of modular forms from theta functions
Advisor : Anupam Saikia

Tezpur University — **India**
Bachelor of Science in Mathematics (with distinction) — *2013-2016*

Research interests

Partitions, q-series, modular forms, special functions, Ramanujan mathematics, experimental mathematics

Articles in preparation

- Congruences for $\mathrm{SOME}(n)$ modulo powers of 5 and 7 via Atkin-Lehner involution (with Aritram Dhar, Frank Garvan and Avi Mukhopadhyay)
- Identities for M_2-rank of partitions without repeated odd parts via Atkin-Lehner involution (with Frank Garvan)
- Rank and crank difference identities modulo p for $p \equiv 3 \pmod 4$ (with Frank Garvan)
- Symmetries for overpartition rank and crank functions (with Frank Garvan)

Publications and preprints

1. Euler's Pentagonal Number Theorem type results via minimal excludants of partitions and overpartitions (with Aritram Dhar, Rajat Gupta and Avi Mukhopadhyay), **submitted**
2. Congruences modulo 3 for sum of certain parts in overpartitions, **submitted**
3. A combinatorial refinement of a sum of parts congruence of Andrews and Dastidar (with Frank Garvan), **submitted**
4. Sum of parts in overpartitions and partitions without repeated odd parts (with Frank Garvan), **submitted** [arXiv]
5. Overpartitions with repeated smallest non-overlined part (with Amita Malik), *Int. J. of Number Theory* (2026), doi:10.1142/S1793042126501411 [DOI] [arXiv]
6. Further analogues of Andrews' spt-function identity, *Bull. Austral. Math. Soc.* (2026), 1–10, doi:10.1017/S0004972726101348 [DOI]
7. On new minimal excludants of overpartitions related to some q-series of Ramanujan (with Aritram Dhar and Avi Mukhopadhyay), *Bull. Austral. Math. Soc.*, **112(1)** (2025), 1-13 [DOI] [arXiv]
8. Bailey pairs, radial limits of q-hypergeometric false theta functions, and a conjecture of Hikami (with Jeremy Lovejoy), *Kyushu J. Math.*, **79** (2025), no. 2, 269–285 [DOI] [arXiv]
9. Transformation and symmetries for the Andrews-Garvan crank function, *Hardy-Ramanujan J.*, **47** (2024), 1-25

[DOI] [arXiv]

10. Combinatorial interpretations of cranks of overpartitions and partitions without repeated odd parts (with Frank Garvan), *SIGMA Symmetry Integrability Geom. Methods Appl.*, SIGMA **20** (2024), 097, 14 pp [DOI] [arXiv]
11. New symmetries for Dyson's rank function (with Frank Garvan), *Ramanujan J.*, **65** (2024), 1883-1939 [DOI] [arXiv]
12. Generalization of the extended minimal excludant of Andrews and Newman (with Aritram Dhar and Avi Mukhopadhyay), *J. Integer Seq.*, **26** (2023), Article No. 23.3.7 [DOI] [arXiv]
13. *Solution to* An application of the Jacobi Triple Product, proposed by G. E. Andrews and M. Merca, Problems and Solutions, *The American Mathematical Monthly*, **130**(8) (2023), 765–778 [DOI]
14. On a Euler-Ramanujan formula, Dirichlet series and minimal surfaces (with Rukmini Dey and Rahul Kumar Singh), *Proc.Indian Acad.Sci.(Math. Sci.)*, **130** (2020), Article No. 61 [DOI] [arXiv]

Awards & Achievements

- **Small scale research grant**, Department of Mathematics, The Pennsylvania State University, Spring 2026
- **Assistant Professor** (tenure-track) job offer (*not availed*) from Florida Gulf Coast University, Fort Myers, Fall 2025
- **CLAS Dissertation Fellowship**, College of Liberal Arts and Sciences, University of Florida, Summer 2025
- **GSC Travel Grant**, Graduate Student Council, University of Florida, Spring 2025
- **Chateaubriand Fellowship**, CNRS and Embassy of France in United States, Fall 2023
- **Chat Yin Ho Award** for outstanding graduate student of the year, Dept. of Math, University of Florida, 2022
- **CLAS Graduate Student Travel Award**, University of Florida, Fall 2022, Spring 2023 & Summer 2024
- **CAM travel grant**, Center for Applied Mathematics, Dept. of Mathematics, University of Florida, Fall 2022
- **Mitacs Globalink Research Award** - for research in Canada, 2018
- **Summer Research Fellowship** of the Indian Academy of Sciences, 2015 & 2016
- **INSPIRE Scholarship** (awarded by Department of Science and Technology, Government of India), 2013-18
- **1st rank** in the State Board examination of 12-th grade, 2013

Research visits

University of Cologne, Germany
Visiting Student *December 15–20, 2023*
Host : Kathrin Bringmann

International Centre for Theoretical Sciences, TIFR, Bangalore, India
Long Term Visiting Student *Aug, 2018–April, 2019*
Research topic : Dirichlet series and Ramanujan identities
Advisor : Rukmini Dey

Simon Fraser University, Burnaby, Canada
SFU-IIT India Initiative Research Intern *May–July, 2018*
Project topic : On a conjecture classifying Littlewood cyclotomic polynomials
Advisor : Stephen Choi

Indian Statistical Institute, New Delhi
Summer Research Fellow *June–July, 2016*
Project Topic : Recurrence sequences and primitive divisors and their application in solving Diophantine equations
Advisor : Shanta Laishram

Indian Statistical Institute, Bangalore
Summer Research Fellow *June–July, 2015*
Project topic : A study on Spectral theorem for normal matrices
Advisor : Jaydeb Sarkar

Talks

Joint Mathematics Meetings (JMM) 2027, Chicago, IL*
AMS Special Session on Partition Theory, q-Series and Related Topics *12–15 Jan, 2027*
"Sum of parts in partitions, overpartitions and partitions without repeated odd parts"

International Congress of Mathematicians (ICM) 2026, Pennsylvania Convention Center, Philadelphia, PA
Poster presentation session *23–30 July, 2026*
"Modular symmetries and identities for rank-type partition statistics"

SIAM Annual Meeting, Cleveland, OH
Mini-symposium on Special Functions with Applications in Number Theory and Combinatorics *6-10 July, 2026*
"Further analogs of Andrews' spt-function identity"

PArtitions and Related Topics I (PART-I), INYAS, India (held online)
"Partition and overpartition rank and crank difference identities modulo p for $p \equiv 3 \pmod 4$" *26–28 June, 2026*

2026 Gainesville International Number Theory conference, University of Florida
Conference in honor of Krishna Alladi's 70th birthday *18-22 March, 2026*
"Overpartition rank and crank differences in terms of Dedekind eta-quotients"

Partitions/combinatorics seminar, Department of Mathematics, Penn State
"Analogs of Andrews' spt-function identity" *28 Oct, 2025*

AMS Fall Southeastern Sectional Meeting, Tulane University, New Orleans, LA
AMS Special Session on Experimental Mathematics *3-5 Oct, 2025*
"Finding identities for rank and crank of partitions and overpartitions using Maple"

Algebra and Number Theory seminar, Department of Mathematics, Penn State
"Modular symmetries for rank-type partition statistics" *23 Sept, 2025*

Partitions/combinatorics seminar, Department of Mathematics, Penn State
"Minimal excludants of partitions and overpartitions" *23 Sept, 2025*

Seminar in Partition Theory, q-series and Related Topics, Michigan Tech (held online)
"Bailey pairs and radial limits of q-hypergeometric false theta functions" *6 Feb, 2025*

Joint Mathematics Meetings (JMM) 2025, Seattle, WA
AMS Special Session on Partition Theory and q-series *8-11 Jan, 2025*
"New symmetries for overpartition rank and crank functions"

Seminar on Number Theory, Tezpur University (held online)
"Bailey pairs and radial limits of q-hypergeometric false theta functions" *19 Nov, 2024*

AMS Fall Central Sectional Meeting, University of Texas at San Antonio, TX
AMS Special Session on Additive Number Theory and Modular Forms *14-15 Sept, 2024*
"Cranks of overpartitions and partitions without repeated odd parts – combinatorial interpretations and mock theta connections"

The Legacy of Ramanujan conference, Penn State University, PA
Poster presentation session *6-9 June, 2024*
"Combinatorial interpretations and mock theta connections of two residual cranks of overpartitions"

Number Theory seminar, Department of Mathematics, University of Florida
"Bailey pairs and radial limits of q-hypergeometric false theta functions" *9 April, 2024*

International conference on modular forms and q-series, University of Cologne
Speed talk session *11-15 March, 2024*
"Pursuing Dyson's dream - unraveling new symmetries for rank-type statistics"

Joint Mathematics Meetings (JMM) 2023, Boston, MA
AMS Special Session on Quadratic Forms, Modular Forms, and Applications *4-7 Jan, 2023*
"New symmetries for the Andrews-Garvan crank function"

SASTRA-Ramanujan Award ceremony and conference, Kumbakonam, India
"New symmetries for rank and crank generating functions" *20-22 Dec, 2022*

AMS Fall Central Sectional Meeting, University of Texas at El Paso, TX
AMS Special Session on The Intersection of Number Theory and Combinatorics *17-18 Sept, 2022*
"Relating the smallest missing part to other partition theoretic objects and statistics"

Number Theory seminar, Department of Mathematics, University of Florida
"Transformation and symmetries for Dyson's rank function" *13 Sept, 2022*

Student Symposium in Combinatorics, UNSW (held online)
"Relating the smallest missing part to other partition statistics" *30 May–3 June, 2022*

47th Annual NY State Graduate Mathematics Conference, Syracause University
"New symmetries for Dyson's rank function" *2 April, 2022*

Masters seminar, Department of Mathematics, Indian Institute of Technology Guwahati
"Cyclotomic polynomials and the Bunyakovski Conjecture" *27 Oct, 2017*

Conferences and workshops attended

The Gross-Zagier Formula, 40 Years Later
Massachusetts Institute of Technology, Cambridge, MA *3-7 Aug, 2026*

Joint Mathematics Meetings (JMM) 2022
held virtually *6-9 April, 2022*

2020 AMS Fall Eastern Sectional Meeting
held virtually, formerly at The Pennsylvania State University *3-4 Oct, 2020*

2019 AMS Fall Southeastern Joint Sectional Meeting
University of Florida, Gainesville *2-3 Nov, 2019*

Advanced Instructional School on Modular forms
Indian Institute of Technology Guwahati *13 May-1 June, 2019*

Algebraic and analytic aspects of automorphic forms
International Centre for Theoretical Sciences, TIFR, Bangalore *25 Feb-7 March, 2019*

Annual Foundation School - I
Indian Institute of Technology Guwahati *Dec 1-31, 2016*

Advanced topics on Discrete Logarithms over Finite Fields and Elliptic Curves
Indian Statistical Institute, New Delhi *14-22 July, 2016*

26th Mid-Year Meeting of scientific lectures of the Indian Academy of Sciences
Indian Institute of Science, Bangalore *3-4 July, 2015*

National Board of Higher Mathematics Workshop on Algebra and Analysis
Tezpur University *17-23 Feb, 2015*

Asian Science Camp 2014
Nanyang Technological University, Singapore *24-29 Aug, 2014*

Referee work

Refereed articles for **International Journal of Number Theory**, **Journal of Algebraic Combinatorics**, **The Electronic Journal of Combinatorics**, **The Journal of the Indian Mathematical Society**, **The Ramanujan Journal** and **Mathematical Reviews (MathSciNet)**

Teaching experience

The Pennsylvania State University

- Instructor for **Calculus with Analytic Geometry I** (MATH 140), Spring 2027
- Instructor for **Matrices** (MATH 220), Fall 2026 & Spring 2026
- Instructor for **Ordinary Differential Equations** (MATH 250), Summer 2026 (online)
- Instructor for **Calculus with Analytic Geometry II** (MATH 141), Spring 2026 & Fall 2025

University of Florida

- Teaching assistant for **Elementary Differential Equations** (MAP 2302), Spring 2025
- Teaching assistant for **Differential Equations for Engineers and Physical Scientists** (MAP 4305) & **Intermediate Differential Equations** (MAP 5304), Spring 2024
- Teaching assistant for **Calculus III** (MAC 2313), Summer 2024 & Fall 2022
- Teaching assistant for **Calculus II** (MAC 2312), Summer 2021 & Spring 2021
- Teaching assistant for **Calculus I** (MAC 2311), Fall 2022, Fall 2020, Spring 2020 & Fall 2019
- Teaching assistant for **Survey of Calculus II** (MAC 2234), Fall 2021
- Teaching assistant for **Survey of Calculus I** (MAC 2233), Fall 2024
- Co-instructor for **Precalculus Algebra** (MAC 1140), Summer 2020
- Instructor for **Precalculus Algebra** (MAC 1140), Summer 2022
- Teaching assistant for **Precalculus Algebra** (MAC 1140), Spring 2022
- Co-instructor for **Basic College Algebra** (MAC 1105), Summer 2023 & Spring 2023

Professional service

- Postdoctoral editor of q-series and friends newsletter (to be published quarterly from December 2026)*
- Co-organizer of Combinatorics/Partitions Seminar in the Department of Mathematics, The Pennsylvania State University, Spring 2026 - present
- Co-organizer of AMS Special Session on Partitions, q-series and modular forms with Ae Ja Yee at the 2026 AMS Fall Eastern Sectional Meeting, Department of Mathematics, George Washington University, Washington, DC, October 3-4, 2026

Graduate coursework

Modern Algebra, Modern Analysis, Functional Analysis, Complex Analysis, Numerical Analysis, Theory of partitions and q-series, Modular Forms and Mock-Theta Functions, Topics in Analytic Number Theory and Sieve Theory, Topics in Combinatorial Theory, Galois Theory, Algebraic Topology, Differential Topology, Probability and Potential Theory

Computer skills

C, C++, LaTeX, Maple, Python, SageMath

References

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*Upcoming